MSC2050 Discrete Mathematics, Presentation 1

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Inha University in Tashkent

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Communication

Ways to communicate:
telegram channel: DM2024IUT
telegram group: DM2024IUT Chat
telegram message: @TomskovaAnna or @Akmalovna_1997
e-mail: a.tomskova@inha.uz

Time to communicate:
from Monday to Saturday
from 9:00 to 18:00
except exam weeks (weeks 8 and 15)

Reasons to communicate:
Any questions on the course is a good reason.

Special instruction:
Any time when you ask question I must know your ID and surname, so each time type them next to the question.

Grading

Midterm: 30 points (offline, during week 8)
Final: 40 points (offline, during week 15)

Homework: 10 points (may get them every 2nd lecture in a week)

Attendance: 10 points (every time you are absent on the lecture you loose 1 point, your behaviour might cost these points)

Quiz: 10 points (every week in e-class, open from 9 am Monday to 9 am next Monday, with average at the end).

Total: 100 points

How to NOT fail the course?

You fail this course (applies to many other courses in IUT) if:

- you are absent at least 9 times during semester (midterms and finals also included in the counter);
- your score less than 40 points in total out of possible 100.
- you cheat on the Final or Midterm; Cheating is
  1. having a mobile phone (on/off, smart/dumb) during examination,
  2. looking around
  3. copying other's work
  4. helping others

These conditions are sufficient but not necessary to fail the course.

How to pass this course?

You should:

- read the textbook in advance and come prepared to the lecture;
- follow the lecture;
- **never** be late or you won't catch up;
- do your **homework** after each lecture;
- write your questions and find out the answers.

This is the only way, unless you took part in High School Mathematics Olympiads on National or possibly International level or know most of the subject already.

Who can help you with your questions?

- The book;
- Internet;
- classmates;
- Mrs Risolat (@Akmalovna_1997)
- Prof Anna (@TomskovaAnna).

Lectures:

- Twice a week (1st lecture is theoretical with Prof Anna, 2nd lecture is practical with Mrs Risolat);
- Attendance is MANDATORY for both lectures;
- We follow the book given in the channel;
- Every Monday in telegram channel I will upload videos from previous online course on DM to HELP you. There are can be some minor changes in syllabus.
- Every Monday I will upload theoretical material for this week (presentation) and HW to do.

Policy on missed classes

- Download and study the presentation files together with the **BOOK** for missed lectures;
- If you have informed me **properly** by an e-mail beforehand or on the same week, and I find your reason to be valid, you will not loose points for attendance;
- If you missed the quiz, come to my office during office hours, I will ask you several questions on the lecture, so you can get 10 points for the quiz. Otherwise, you get 0 points.

  Dear Dr. Tomskova,
  I cannot attend the tutorials and classes on 17th of October
  due to being hospitalized with an acute appendicitis
  from 15th till 18th. Attached is a picture of a doctor's note.

  Best regards,
  Student of CSE-17-1, U17099999,
  John Doe

Course content

Before Midterm, the following topics for the Midterm exam:

- **Logic** (Chapter 1)
- **Naive set theory.** *Some Theoretical Computer Science* (Chapter 2)
- **Algorithms** (Chapter 3)
- **Introductory Number Theory and Cryptography** (Chapter 4)

After Midterm, the following topics for the Final Examination

- **Induction and Recursion** (Chapter 5)
- **Relations** (Chapter 9)
- **Graphs** (Chapter 10)
- **Trees** (Chapter 11)

Some English vocabulary

‘Discreet’ or ‘discrete’?

- from the Latin *discretus*, meaning ‘separate’;
- are pronounced the same way;
- but have different meanings.
- Discrete stays true to its Latin origins and means ‘individually separate and distinct’, as in:

  This issue is discrete from the others.
- Discreet means 'careful and circumspect', as in:

  A lot of their work is carried out in a very discreet and confidential manner.
- *Circumspect* – wary and unwilling to take risks.

  The officials were very circumspect in their statements.

In Mathematics, Physics and Computer Sciences

Discrete is not continuous. It is related to being finite, or countable.

DM is the mathematics behind CS

- modeling computing structures
- designing programs and algorithms
- reasoning about programs and algorithms
- solving real-world problems using computer

Aspects of discrete mathematics:

1. logical thinking (formal logic, symbolic manipulation of notation, logic in programming, logic in circuits)
2. relational thinking (sets, functions, relations, partial ordering, graphs)
3. recursive thinking (recurrence relations and definitions, proofs by induction, recursive data structures, recursive algorithms)
4. quantitative thinking (counting, combinations, permutations, arrangements, the pigeonhole principle, estimating growth of functions, big-Oh notation)
5. analytical thinking (writing programs that are correct, writing algorithms that are efficient)
6. applied thinking (making the bridge between the mathematical tools and problems we need to solve).

Propositional Logic

Propositions

A proposition is a declarative *sentence* that is either true or false, but not both.

Examples of propositions

- London is the capital of Great Britain. (TRUE)
- Sydney is the capital of Australia. (FALSE)
- 1+1=2 (TRUE)
- 2+2=5 (FALSE)

Example (the following are not propositions)

- What time is it?
- Read this carefully.
- $x^2 + y^2 = z^2$.
- Wow.

Answer Q1

- Denote propositions with conventional letters $p, q, r, s, \ldots$. These are propositional **variables** .
- The *truth value* of a proposition is true, if it is a true proposition. It is denoted by T or 1.
- The *truth value* of a proposition is false, if it is a false proposition. It is denoted by F or 0.
- The area of logic that deals with propositions is called the *propositional calculus* or ***propositional logic***.
- It was first developed systematically by the Greek philosopher Aristotle more than 2300 years ago.
- New propositions, called *compound propositions*, are formed from existing propositions using logical operators.
- We follow the methods of the book "The Laws of Thought" by an English mathematician George Boole, 1854.

Negation

Let p be a proposition. The **negation** of p is

- the statement "It is not the case that p"
- denoted by $\neg p$
- the proposition $\neg p$ is read "not p"
- The truth value of $\neg p$, is the opposite of the truth value of p

p	$\neg p$
0	1
1	0

p="Students have at least 6 courses per semester". Then $\neg p$ is

"It is not the case that students have at least 6 courses per semester"

or "Students have no more than 6 courses per semester"

Answer Q2

Conjunction

Let p and q be propositions. The **conjunction** of p and q is

- the proposition "p and q"
- denoted by $p \wedge q$
- The conjunction $p \wedge q$ is true when both p and q are true and is false otherwise.
- In logic the word "but" sometimes is used instead of "and" in a conjunction:
  "The sun is shining, but it is raining " $= s \wedge r$,
  where s="the sun is shining", r="it is raining"

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Answer Q3

Disjunction

Let p and q be propositions. The **disjunction** of p and q is

- the proposition "p or q"
- denoted by $p \vee q$
- The disjunction $p \vee q$ is false when both p and q are false and is true otherwise.

Exclusive or (XOR)

The exclusive or of p and q, denoted by $p \oplus q$, is the proposition that is true when exactly one of p and q is true and is false otherwise.

p	q	$p \vee q$	$p \oplus q$
T	T	T	F
T	F	T	T
F	T	T	T
F	F	F	F

Answer Q4

Conditional Statement*

Let p and q be propositions.

- The *conditional statement* $p \to q$ is the proposition "if p, then q".
- The conditional statement $p \to q$ is false when p is true and q is false, and true otherwise.
- p is called the hypothesis (or antecedent or *premise*)
- q is called the *conclusion* (or consequence).

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

If you score less than 40 points, then you will fail DM. $[T \to T = T]$
If you score more than 95 points, then you will get A+. $[T \to F = F]$
If I'm the king of Wakanda, then you are a superhero. $[F \to F = T]$
If it is Engineering Mathematics class, then you are from SOCIE. $[F \to T = T]$
Answer Q5

Other terminology for $p \to q$ (reminder $T \to F = F$)

if p, then q	q follows from p
if p, q	q when p
p implies q	q whenever p
p $\boxed{\text{only if}}$ q	q if p
p is sufficient for q	q is necessary for p
a necessary condition for p is q	a sufficient condition for q is p
	q unless $\neg p$

If **Maria learns discrete mathematics**, then *she will find a good job*.

Maria will find a good job when **she learns discrete mathematics**.

For *Maria to get a good job,* it is sufficient for **her to learn discrete mathematics**.

Maria will find a good job unless **she does not learn discrete mathematics**.

Other terminology for $p \to q$ (reminder $T \to F = F$)

if p, then q	q follows from p
if p, q	q when p
p implies q	q whenever p
p $\boxed{\text{only if}}$ q	q if p
p is sufficient for q	q is necessary for p
a necessary condition for p is q	a sufficient condition for q is p
	q unless $\neg p$

To say "p only if q" means that p can only ever be true when q is true. That is, q is necessary for p to be true.

I will find a million dollars inside this locker only if I know the combination.

But that doesn't mean I will find a million dollars there if I know the combination. After all, there might be only 15 thousands Uzbek soms in there.

Other terminology for $p \to q$ (reminder $T \to F = F$)

if p, then q	q follows from p
if p, q	q when p
p implies q	q whenever p
p $\boxed{\text{only if}}$ q	q if p
p is sufficient for q	q is necessary for p
a necessary condition for p is q	a sufficient condition for q is p
	q unless $\neg p$

You can have dessert only if *you finish your meal*.
For **you to have a dessert** it is sufficient *to finish your meal*.

However, it is not this: "If *you finish your meal* then **you can have a dessert** ".

"If it is sunny, then we will go to the beach" is not a conditional statement by its meaning...What is meant is if and only if.

Converse, contrapositive and inverse

Given a proposition $p \to q$

- The proposition $q \to p$ is called the converse of $p \to q$.
- The contrapositive of $p \to q$ is the proposition $\neg q \to \neg p$.
- The proposition $\neg p \to \neg q$ is called the inverse of $p \to q$.

p	q	$p \to q$	$q \to p$	$\neg q \to \neg p$	$\neg p \to \neg q$
T	T				
T	F	F		F	
F	T		F		F
F	F				

p	q	$p \to q$	$q \to p$	$\neg q \to \neg p$	$\neg p \to \neg q$
1	1	1	1	1	1
1	0	0	1	0	1
0	1	1	0	1	0
0	0	1	1	1	1

Conclusion: $p \to q$ and $\neg q \to \neg p$ have the same truth table.
A proposition and its contrapositive are the same.

Biconditional

$p \leftrightarrow q$ which is "p if and only if q"

The biconditional statement $p \leftrightarrow q$ is true when p and q have the same truth values, and is false otherwise. Biconditional statements are also called bi-implications.

p	q	$p \leftrightarrow q$	$p \to q$	$q \to p$	$(p \to q) \wedge (q \to p)$
1	1	1	1	1	1
1	0	0	0	1	0
0	1	0	1	0	0
0	0	1	1	1	1

Hence, $(p \to q) \wedge (q \to p)$ and $p \leftrightarrow q$ have the same truth table.
Example: You can take the flight if and only if you buy a ticket.
Other ways to express $p \leftrightarrow q$

- p is necessary and sufficient for q
- If p, then q, and conversely.
- p iff q

Answer Q6

Precedence of Logical Operations

Operator	Precedence
1	$\neg$
2	$\wedge$
3	$\vee$
4	$\rightarrow$
5	$\leftrightarrow$

Example: $p \rightarrow q \vee \neg r = p \rightarrow (q \vee (\neg r))$
If we want implication to take precedence over disjunction we use parenthesis $(p \rightarrow q) \vee \neg r$

Homework:
- p. 11 of the book read the paragraph "Logic and Bit operations"
- Solve all *odd* exercises at p. 12 (there are answers in the end of the book for odd numbers)

Applications of Propositional Logic

Translation into logic

You can access the Internet from campus only if you are a computer science major or you are not a freshman.

- $a =$ you can access the Internet from campus
- $c =$ you are a computer science major
- $f =$ you are a freshman
- a only if (c or not f)
- $a \to (c \vee \neg f)$

You cannot ride the roller coaster if you are under 4 feet tall unless you are older than 16 years old.

- *If you are NOT older than 16 years old* then **you cannot ride the roller coaster if you are under 4 feet tall**.
- $\neg OLD \to (short \to \neg R)$

Answer Q7

System Specifications

System and software engineers take requirements in natural language and produce precise and unambiguous specifications that can be used as the basis for system development.

The automated reply cannot be sent when the file system is full.

The file system is full $= f$

The automated reply can be sent $= s$.

Then the statement is

$$\text{"}\neg s \text{ when } f\text{"}$$

which is $f \to \neg s$.

Consistency of System Specifications

System specifications are **consistent** if they do not contain conflicting requirements.
To check that system specifications is consistent there is must be situation when all system specifications are simultaneously true.
Example: Is the following system specifications are consistent.
"If the file system is not locked, then new messages will be queued"
"If new messages are not queued, then they will be sent to the buffer."
"If the file system is not locked, then new messages will be sent to the buffer"
"New messages will not be sent to the buffer"

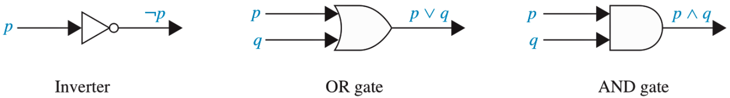
$$\begin{aligned} &\neg l \to q && (1)\\ &\neg q \to b && (2)\\ &\neg l \to b && (3)\\ &\neg b && (4) \end{aligned}$$

(4)=1 implies $b = 0$
(3)=1 implies $\neg l = 0$ implies $l = 1$
(2)=1 implies $\neg q = 0$ implies $q = 1$
(1) is true, since $0 \to 1$ is true

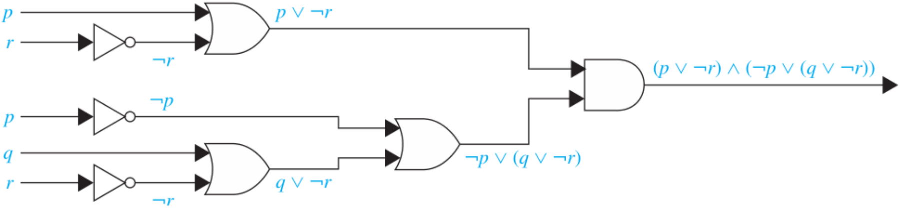
So the system specifications are consistent.
Answer Q8

Logic Circuits

Propositional logic can be applied to the design of computer hardware. This was first observed in 1938 by Claude Shannon in his MIT master's thesis.



Suppose that we have a formula for the output of a digital circuit in terms of negations, disjunctions, and conjunctions. Then, we can systematically build a digital circuit with the desired output.



Homework:

- p. 18 of the book read the paragraph "Boolean Searches"
- Solve all *odd* exercises at p. 22, #1-#13, #41 and #43.

For someone interested (not necessarily)

- p. 19 of the book read the paragraph "Logic Puzzles" and solve #15-#39.

Propositional Equivalences

Tautology and Contradiction

A compound proposition that is always true is called a **tautology**.
Tautology is denoted by **T** or **1**.
Proposition $p \vee \neg p$ is a tautology.

p	$\neg p$	$p \vee \neg p$
1	0	1
0	1	1

A compound proposition that is always false is called a **contradiction**, denoted by **F**, or **0**.
Proposition $p \wedge \neg p$ is a contradiction.

Answer Q9

A compound proposition that is neither a tautology nor a contradiction is called a **contingency**.

Equivalent propositions

Definition

The compound propositions p and q are called *logically equivalent* if $p \leftrightarrow q$ is a tautology. The notation $p \equiv q$ denotes that p and q are logically equivalent.

Note that, $p \leftrightarrow q$ is tautology if and only if compound proposition p and q are simultaneously true or simultaneously false, which means
their truth tables are the same!

For example, as we have just seen

p	$\neg p$	$p \vee \neg p$	**T**
1	0	1	1
0	1	1	1

So we write $p \vee \neg p \equiv$ **T**.

Similarly, $p \wedge \neg p \equiv$ **F**.

De Morgan's Law

Negation of "Heather will go to the concert **or** Steve will go to the concert" is "Heather will **not** go to the concert **and** Steve will **not** go to the concert".

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

p	g	$p \vee g$	$\neg(p \vee g)$	$\neg p$	$\neg g$	$\neg p \wedge \neg g$
1	1	1	0	0	0	0
1	0	1	0	0	1	0
0	1	1	0	1	0	0
0	0	0	1	1	1	1

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

Negation of "Miguel has a cellphone **and** he has a laptop computer" is "Miguel does **not** have a cellphone **or** he does **not** have a laptop computer".

Answer Q10

Elementary Logical Equivalences

- $p \wedge \mathbf{T} \equiv p, \ p \vee \mathbf{F} \equiv p$ — Identity laws
- $p \vee \mathbf{T} \equiv \mathbf{T}, \ p \wedge \mathbf{F} \equiv \mathbf{F}$ — Domination laws
- $p \vee p \equiv p, \ p \wedge p \equiv p$ — Idempotent laws
- $\neg(\neg p) \equiv p$ — Double negation law
- $p \vee q \equiv q \vee p, \ p \wedge q \equiv q \wedge p$ — Commutative laws
- $(p \vee q) \vee r \equiv p \vee (q \vee r)$
  $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ — Associative laws
- $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
  $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ — Distributive laws
- $\neg(p \wedge q) \equiv \neg p \vee \neg q$
  $\neg(p \vee q) \equiv \neg p \wedge \neg q$ — De Morgan's laws
- $p \vee (p \wedge q) \equiv p$
  $p \wedge (p \vee q) \equiv p$ — Absorption laws
- $p \vee \neg p \equiv \mathbf{T}, \ p \wedge \neg p \equiv \mathbf{F}$ — Negation laws

Proof of Distributive law using truth table

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
1	1	1	1	1	1	1	1
1	1	0	0	1	1	1	1
1	0	1	0	1	1	1	1
1	0	0	0	1	1	1	1
0	1	1	1	1	1	1	1
0	1	0	0	0	1	0	0
0	0	1	0	0	0	1	0
0	0	0	0	0	0	0	0

Proof of the second De Morgan's law using the first one and the negation law

$$\begin{aligned} \neg(p \vee q) &\equiv \neg(\neg(\neg p) \vee \neg(\neg q)) && \textit{by the double negation law} \\ &\equiv \neg\neg(\neg p \wedge \neg q) && \textit{by the first De Morgan law} \\ &\equiv \neg p \wedge \neg q && \textit{by the double negation law} \end{aligned}$$

Logical equivalences involving conditional statements

1. $\boxed{p \to q \equiv \neg p \vee q}$
2. $p \to q \equiv \neg q \to \neg p$ *(rule: proposition is equivalent to its contrapositive)*
3. $p \vee q \equiv \neg p \to q$ *(follows from 1 if you change p to $\neg p$)*
4. $p \wedge q \equiv \neg(p \to \neg q)$
5. $\neg(p \to q) \equiv p \wedge \neg q$ *(negate 1 and use De Morgan's Law and 1 again)*
6. $(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$
7. $(p \to q) \vee (p \to r) \equiv p \to (q \vee r)$
8. $(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$
9. $(p \to r) \vee (q \to r) \equiv (p \wedge q) \to r$

All of the equivalences can be established by constructing the truth table. Alternatively, one can prove it using (1) and applying laws of conjunction, disjunction and negation.

Examples

Show that $\neg(p \to q)$ and $p \wedge \neg q$ are logically equivalent.

Solution

$$\begin{aligned}
\neg(p \to q) &\equiv \neg(\neg p \vee q) && \textit{by Law 1 on implications} \\
&\equiv \neg(\neg p) \wedge \neg q && \textit{by the second De Morgan law} \\
&\equiv p \wedge \neg q && \textit{by the double negation law}
\end{aligned}$$

Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent.

Solution

$$\begin{aligned}
\neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) && \textit{by the second De Morgan law} \\
&\equiv \neg p \wedge [\neg(\neg p) \vee \neg q] && \textit{by the first De Morgan law} \\
&\equiv \neg p \wedge (p \vee \neg q) && \textit{by the double negation law} \\
&\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) && \textit{by the second distributive law} \\
&\equiv \mathbf{F} \vee (\neg p \wedge \neg q) && \textit{because } \neg p \wedge p \equiv \boldsymbol{F} \\
&\equiv (\neg p \wedge \neg q) \vee \mathbf{F} && \textit{Commutative law} \\
&\equiv \neg p \wedge \neg q && \textit{by the identity law for F}
\end{aligned}$$

Answer Q11

Normal form

p	q	r	$p \land q \land r$
1	1	1	**1**
1	1	0	0
1	0	1	0
1	0	0	0
0	1	1	0
0	1	0	0
0	0	1	0
0	0	0	0

Considering the input (1-1-1) we create the **normal** form:

$$p \land q \land r.$$

Note that the truth table of $p \land q \land r$ has only one true value if $p = q = r = 1$ (it is zero otherwise).

Note that the truth table of $p \land \neg q \land r$ has only one true value if $p = r = 1$ and $q = 0$.

Note that the truth table of $\neg p \land \neg q \land r$ has only one true value if $r = 1$ and $p = q = 0$.

and so on....

Answer Q12

Disjunctive normal form

p	q	r	?
1	1	1	**1**
1	1	0	0
1	0	1	0
1	0	0	**1**
0	1	1	0
0	1	0	0
0	0	1	**1**
0	0	0	0

We want to construct a compound proposition for a given truth table!

Observe that for $p \vee q \vee r$ to have true value it is enough that one of p, q, r has true value.

The unknown proposition of the table is then

$$(p \wedge q \wedge r) \vee (p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r).$$

That disjunction of normal forms is called disjunctive normal form. Given any truth table, we can construct a proposition that corresponds to that table. Moreover, any proposition admits a disjunctive normal form (build a truth table and take disjunction of the normal forms corresponding to the truth values).

Any proposition is logically equivalent to a compound proposition involving only $\vee, \wedge, \neg$ logical operators. Any proposition admits its logical circuit.

Answer Q13

Find the disjunctive normal form of a proposition $p \to (q \wedge r)$.

Solution: By building the truth table, observe that there are 5 ones and the following is built:

$$(p \wedge q \wedge r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r) \vee (\neg p \wedge \neg q \wedge \neg r).$$

Another method without constructing the truth table

$$\begin{aligned}
p \to (q \wedge r) &\equiv \neg p \vee (q \wedge r) \equiv (\neg p \wedge 1 \wedge 1) \vee (1 \wedge q \wedge r) \\
&\equiv (\neg p \wedge (q \vee \neg q) \wedge (r \vee \neg r)) \vee ((p \vee \neg p) \wedge q \wedge r) \\
&\equiv (\neg p \wedge (q \vee \neg q) \wedge (r \vee \neg r)) \vee ((p \vee \neg p) \wedge (q \wedge r)) \\
&\equiv (\neg p \wedge (q \vee \neg q) \wedge (r \vee \neg r)) \vee (((p \wedge (q \wedge r)) \vee (\neg p \wedge (q \wedge r))) \\
&\equiv (\neg p \wedge (q \vee \neg q) \wedge (r \vee \neg r)) \vee (p \wedge q \wedge r) \vee (\neg p \wedge q \wedge r) \qquad (1)
\end{aligned}$$

Using distributivity, by analogue one obtains that $(\neg p \wedge (q \vee \neg q) \wedge (r \vee \neg r)) \equiv$ $(\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r) \vee (\neg p \wedge \neg q \wedge \neg r)$. (2)

The result follows from substituting (2) into (1).
Question: Why do we have 6 normal forms, while in the answer there are only 5?

Propositional Satisfiability

A compound proposition is **satisfiable** if there is an assignment of truth values to its variables that makes it true.

When the compound proposition is false for all values of its variables, the proposition is **unsatisfiable**.

When we find a particular assignment of values that makes a compound proposition true, we have shown that it is satisfiable.

Is the proposition $p \to (q \wedge r)$ satisfiable?

Solution: By constructing its disjunctive normal form

$$(p \wedge q \wedge r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r) \vee (\neg p \wedge \neg q \wedge \neg r).$$

we see that the proposition is true for five assignment of values (write them).

Homework:

- p. 32 of the book read the paragraph "Applications of Satisfiability"
- Solve all *odd* exercises starting at p. 34 till #61.

For someone interested (not necessarily)

- p. 36 solve #63-#66.